# THE HENSELIZATION OF A PAIR DEPENDS ONLY ON THE RADICAL OF THE IDEAL

THE STACKS PROJECT AUTHORS

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*Allowed standard prerequisites (editorial).*

- The constructed pair is henselian with respect to the extended ideal of the construction. Lemma \ref{lemma-henselization-flat}, cited at proof lines 3221-3222. Identifies the ideals of the two constructed pairs.
- Replacing the ideal by another ideal with the same vanishing locus preserves the henselian property. Lemma \ref{lemma-change-ideal-henselian-pair}, cited at proof lines 3223-3225. Makes each constructed ring henselian with respect to the other ideal.
- The universal property of the construction of the henselization of a pair. Lemma \ref{lemma-henselization}, cited at proof lines 3226-3227. Produces the two ring maps and their uniqueness.

*Source reference.* Original lemma 0F0L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18766–18766. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 0A02: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12620–12620. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0AGU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13207–13207. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 09XJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12532–12532. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $A$ be a ring with ideals $I$ and $J$. If $V(I) = V(J)$ then the functor of Lemma 0A02 produces the same ring for the pair $(A, I)$ as for the pair $(A, J)$.*

*Proof.* Let $(A', IA')$ be the pair produced by Lemma 0A02 starting with the pair $(A, I)$, see Lemma 0AGU. Let $(A'', JA'')$ be the pair produced by Lemma 0A02 starting with the pair $(A, J)$. By Lemma 09XJ we see that $(A', JA')$ is a henselian pair and $(A'', IA'')$ is a henselian pair. By the universal property of the construction we obtain unique $A$-algebra maps $A'' \to A'$ and $A' \to A''$. The uniqueness shows that these are mutually inverse. □